PU5558-PU5567-computer-lab-report

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Introduction

The following report is based on Provisional Patient Reported Outcome Measures (PROMS) specifically on the outcome of Hip and Knee Replacement procedures completed between April 2020 to March 2021, provided by NHS England. For this report the focus will be on the outcome measures of knee replacement surgery. Specific outcome measures pertaining to the Quality Of Life (QOL) to the patient have been recorded through a visual analouge scale designed to measure the self-reported status of the patients quality of life pre/post surgery. These values gathered from the survery will be used to inform the models.

Load necessary packages

```
library(tidyverse)
library(tidymodels)
library(corrplot)
library(psych)
library(ggplot2)
library(rmarkdown)
library(dplyr)
library(reshape2)
library(vip)
library(Hmisc)
library(VIM)
library(caret)
library(flextable)
library(gtsummary)
library(data.table)
library(randomForest)
```

Load chosen dataset

```
kneeData <- read_csv("Knee_Replacement_CCG 2021.csv")
kneeData <- kneeData %>%
    select(-Procedure, -Year, -"Revision Flag", -"Provider Code") %>%
    drop_na() %>%
    unique()
kneeDataNoImpute <- kneeData
```

```r
glimpse(kneeData)
```

```
## Rows: 4,528
## Columns: 77
## $ `Age Band`                                      <chr> "*", "*", "*", "*",~
## $ Gender                                          <chr> "*", "*", "*", "*",~
## $ `Pre-Op Q Assisted`                             <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Assisted By`                          <dbl> 0, 0, 0, 0, 0, 0, 0~
## $ `Pre-Op Q Symptom Period`                       <dbl> 2, 4, 4, 2, 4, 2, 2~
## $ `Pre-Op Q Previous Surgery`                     <dbl> 2, 1, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Living Arrangements`                  <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Disability`                           <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Heart Disease`                                 <dbl> 9, 9, 9, 9, 9, 1, 9~
## $ `High Bp`                                       <dbl> 1, 9, 9, 1, 1, 1, 9~
## $ Stroke                                          <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Circulation                                     <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Lung Disease`                                  <dbl> 9, 9, 9, 1, 9, 9, 9~
## $ Diabetes                                        <dbl> 9, 9, 9, 1, 9, 9, 9~
## $ `Kidney Disease`                                <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Nervous System`                                <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Liver Disease`                                 <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Cancer                                          <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Depression                                      <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Arthritis                                       <dbl> 9, 1, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Mobility`                             <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Self-Care`                            <dbl> 1, 1, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Activity`                             <dbl> 3, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Discomfort`                           <dbl> 2, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Anxiety`                              <dbl> 1, 1, 2, 2, 1, 2, 1~
## $ `Pre-Op Q EQ5D Index Profile`                   <dbl> 21321, 21221, 21332~
## $ `Pre-Op Q EQ5D Index`                           <dbl> 0.364, 0.691, 0.030~
## $ `Post-Op Q Assisted`                            <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Assisted By`                         <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Post-Op Q Living Arrangements`                 <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Post-Op Q Disability`                          <dbl> 1, 2, 2, 2, 2, 1, 2~
## $ `Post-Op Q Mobility`                            <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Self-Care`                           <dbl> 2, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Activity`                            <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Discomfort`                          <dbl> 1, 1, 1, 1, 1, 2, 2~
## $ `Post-Op Q Anxiety`                             <dbl> 2, 1, 1, 1, 1, 1, 1~
## $ `Post-Op Q Satisfaction`                        <dbl> 2, 3, 1, 2, 1, 4, 3~
## $ `Post-Op Q Sucess`                              <dbl> 1, 1, 1, 1, 1, 4, 1~
## $ `Post-Op Q Allergy`                             <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Bleeding`                            <dbl> 2, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Wound`                               <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Urine`                               <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Further Surgery`                     <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Readmitted`                          <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q EQ5D Index Profile`                  <dbl> 12112, 11111, 11111~
## $ `Post-Op Q EQ5D Index`                          <dbl> 0.744, 1.000, 1.000~
## $ `Knee Replacement EQ 5D Index Post-Op Q Predicted` <dbl> 0.6621424, 0.740953~
## $ `Pre-Op Q EQ VAS`                               <dbl> 85, 50, 46, 60, 60,~
## $ `Post-Op Q EQ VAS`                              <dbl> 60, 100, 90, 95, 90~
## $ `Knee Replacement EQ VAS_Post-Op Q Predicted`   <dbl> 75.63073, 66.38606,~
```

```
## $ `Knee Replacement Pre-Op Q Pain`                  <dbl> 1, 1, 0, 0, 1, 1, 1~
## $ `Knee Replacement Pre-Op Q Night Pain`            <dbl> 1, 2, 2, 2, 2, 3, 0~
## $ `Knee Replacement Pre-Op Q Washing`               <dbl> 3, 4, 3, 4, 3, 3, 3~
## $ `Knee Replacement Pre-Op Q Transport`             <dbl> 1, 4, 2, 4, 2, 2, 2~
## $ `Knee Replacement Pre-Op Q Walking`               <dbl> 3, 3, 2, 3, 3, 1, 2~
## $ `Knee Replacement Pre-Op Q Standing`              <dbl> 3, 2, 2, 3, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Limping`               <dbl> 0, 3, 0, 0, 3, 1, 1~
## $ `Knee Replacement Pre-Op Q Kneeling`              <dbl> 0, 0, 0, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Work`                  <dbl> 2, 2, 1, 3, 2, 1, 2~
## $ `Knee Replacement Pre-Op Q Confidence`            <dbl> 3, 3, 3, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Shopping`              <dbl> 2, 4, 0, 2, 3, 0, 2~
## $ `Knee Replacement Pre-Op Q Stairs`                <dbl> 1, 4, 2, 2, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Score`                 <dbl> 20, 32, 17, 27, 27,~
## $ `Knee Replacement Post-Op Q Pain`                 <dbl> 4, 3, 3, 4, 3, 1, 2~
## $ `Knee Replacement Post-Op Q Night Pain`           <dbl> 4, 4, 3, 4, 4, 3, 2~
## $ `Knee Replacement Post-Op Q Washing`              <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Transport`            <dbl> 2, 4, 4, 4, 4, 3, 4~
## $ `Knee Replacement Post-Op Q Walking`              <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Standing`             <dbl> 3, 4, 4, 4, 4, 1, 3~
## $ `Knee Replacement Post-Op Q Limping`              <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Kneeling`             <dbl> 0, 4, 0, 3, 2, 0, 2~
## $ `Knee Replacement Post-Op Q Work`                 <dbl> 3, 3, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Confidence`           <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Shopping`             <dbl> 4, 4, 4, 4, 4, 0, 4~
## $ `Knee Replacement Post-Op Q Stairs`               <dbl> 3, 4, 4, 4, 3, 1, 4~
## $ `Knee Replacement Post-Op Q Score`                <dbl> 37, 46, 42, 47, 44,~
## $ `Knee Replacement OKS Post-Op Q Predicted`        <dbl> 34.02619, 36.88715,~
```

```r
# glimpseData <- kneeData[, c("Age Band", "Gender", "Pre-Op Q Assisted", "Heart Disease", "Arthritis",
#       "Post-Op Q Bleeding", "Pre-Op Q EQ5D Index Profile", "Post-Op Q EQ5D Index Profile",
#       "Pre-Op Q EQ5D Index", "Post-Op Q EQ5D Index")]
# kable(glimpseData, format = "latex", booktabs = TRUE, longtable = TRUE,
#     caption="Rate of Change within an Individual Health Board Against its Moving Average") %>%
#     kable_styling(latex_options = c("striped", "hold_position")) %>%
#     row_spec(0, bold = TRUE) %>%
#     column_spec(1, "2cm") %>%
#     row_spec(15, bold = TRUE) %>%
#     kableExtra::landscape()

# set_flextable_defaults(
#     font.font.family = "Arial", font.size = 10,
#     border.color = "gray", big.mark = ""
# )


# ft <- flextable(head(kneeData), col_keys = c(
#     "Age Band", "Gender", "Pre-Op Q Assisted", "Heart Disease", "Arthritis",
#     "Post-Op Q Bleeding", "Pre-Op Q EQ5D Index Profile", "Post-Op Q EQ5D Index Profile",
#     "Pre-Op Q EQ5D Index", "Post-Op Q EQ5D Index"
# )) |> bold(part = "header")
# set_table_properties(ft, width=.5, layout = "autofit")
```

Dataset description

This dataset uses data from NHS England regarding PROMS for knee replacement surgery. The way in which this data has been gathered is to collect self-reported symptoms/QOL metrics through two standard procedures. One survey is the EQ-5DTM Index designed to evaluate the QOL in terms of of doing daily tasks,such as(cite NHS PROMS): * Ability to pursue usual activites * Current experience of anxiety and/or depression, if any * Current experience of pain and discomfort if any * Mobility * Ability to wash and dress themselves The closer the EQ5 value is to 1, the better the patients QOL.

While the EQ5 focuses on day to day tasks the EQ-VAS focuses on pre/post operative outcomes through a visual thermomemter scale rannging from 0 to 100. This scale is designed to aid patients in describing how they feel post operation about the results and if their problems pre operation have been addressed with the surgery.

Visually inspecting the database we see that Age Band and Gender have many missing values. Alongside this, there are many missing values for co-morbidites. As instructed all missing values for co-morbidites denotes as 9 will be taken as a no value. However, since Age Band and Gender typically are important factors (especially as confounders), these values will be imputed. Since there are a possible 77 variables to provide summary statstics on, only specific columns relevant to the following questions below will be considered.

```r
kneeData |> tbl_summary(include = c(
    "Age Band", "Gender", "Knee Replacement Post-Op Q Pain",
    "Post-Op Q EQ5D Index", "Pre-Op Q Previous Surgery", "Post-Op Q Assisted By"
))
```

Suitable machine learning algorithm for three questions:

1. Before the operation, can we estimate the post-operative EQ5D index for a patient? The EQ5 index is provided as a float value between 0 and 1. Due to its continous nature a suitable supervised model to answer this question would be a linear regression model. Since we wish to predict a continous variable against possible predictor variables pre/post surgery, we would have to find correlations between an increasing EQ5 index value and variables in the dataset. There are 2 possible approaches to answering this question: a simple linear regression model and a random forest for regression. Both models wont be made, however assumptions on the dataset will guide the choice of model; if the dataset is found to have a non-linear relationship or standard test to check for normality does not hold true, a random forest will be used. A linear regression model is simpler to run and provides better interpratibility of the model, however if assumptions do not hold, a random forest for regression must be used. This question will be used to complete the report.

2. Before the operation, can we predict how much pain a patient will have after the operation? Since pain is measured as ordinal categories, a model designed for classification would fit this specification. Referencing (mention table figure) we see that 25% of data points are missing values for the questions regarding pain. If we were to use an unsupervised model it may not capture the true relationship between predictors and the outcome variable due to the vast number of unlabelled data points for a siginificant predictor. To tackle this issue of missing data, a semi-supervised model such as a semi-supervised support vector machine can overcome the missing data barrier. The semi-supervised SVM would use a mixture of labelled an unlabelled data to create descion boundaries for the different pain categories, and predict the positioning of a data point to its relative class given the presence of certain predictor variables. The level of computation a semi-supervised SVM has may be determinal to the models overall performance, but coupled with the fact that there are many other missing values in other questions regarding the QOL status of the patient, imputation on specific question would be difficult to complete without clinical knowledge to inform the possible current state of the patient. Therefore, the increased computational power for the model may be benefical when trying to address the gaps in data.

3. Before the operation, can we calculate how many patients have had previous surgery?

Here we can use supervised/semi-supervised learning methods to cualcyate the numerical value of the patiens that have had the previous surgery. The results is binary wether they have or have not had it

Characteristic	**N = 4,528**[1]
Age Band	
*	1,603 (35%)
50 to 59	164 (3.6%)
60 to 69	1,034 (23%)
70 to 79	1,511 (33%)
80 to 89	216 (4.8%)
Gender	
*	1,603 (35%)
1	1,326 (29%)
2	1,599 (35%)
Knee Replacement Post-Op Q Pain	
0	116 (2.6%)
1	773 (17%)
2	974 (22%)
3	1,529 (34%)
4	1,136 (25%)
Post-Op Q EQ5D Index	0.76 (0.69, 1.00)
Pre-Op Q Previous Surgery	
1	437 (9.7%)
2	4,060 (90%)
9	31 (0.7%)
Post-Op Q Assisted By	
1	252 (5.6%)
2	18 (0.4%)
3	6 (0.1%)
4	9 (0.2%)
5	2 (<0.1%)
6	9 (0.2%)
9	4,232 (93%)

[1] n (%); Median (Q1, Q3)

Model building to answer chosen question

1. Data splitting

Before the data is split into relevant train/test splits. The issue of Age Band and Gender both missing 35% of their data needs to be a handled. The following imputation is simple and doesnt use external questions or factors to inform descion making. Age Band is recoded into group numbers and the median of the group numbers is taken. Then the median value is then imputed to the missing data.

Imputation of the Gender column is slightly more complex, with the method being a k Nearest Neighbour (kNN) approach. This approach tries to maintain the randomness of the gender data points, (reference table) looking at the table there are 6% more females that get the surgery than males. Therefore using a simple 50-50 or a median split would lose the close difference in value between the different genders. The KNN maintian thre nuisance in this split. The following table summaries the changes in the data values after imputation, the changes are as follows * Age Band - Median = Category 3. Increase of 35% or 1,603 patients in category 3 * Gender - Male: Increase of 693 cases (+26%), Female: Increase of 910 cases (+20%)

Replacing missing values was only complete for age and gender since these values can be more easily accounted for in comparison to missing questionare values. Replacing missing questionare values may have serious consequences to the predictions of the model, creating false equivalences. With certain predictors given more importance due to the inputted data. Removing these data values was not an option, since we would be losing up to 80% of the questionare data with some questions (hilight table). To ensure that the integrity of the the patient reported outcomes remain the same, more emphasis will be placed on the models to handle missing data such as a random forest (RF) model for regression. RF models dont need imputation of missing values and they dont suffer from underpreforming from the lack of labelled data points. This is due to their automatic handiling of missing data through surrogate splits. The randomForest package in R states in their documentation that imputation and handiling of missing values will only take place for values coded as NA, since the missing values in this dataset are coded as 9, these will be changed.

As per the assignment specifications, 9 will be handled as "No" for any columns pertaining to a previous diagnosis of condition. 9 will be recoded to 2 for these answers. The randomForest package comes with its own built in imputation algorithim, based on first using a median/mean as a baseline to then create its own forests based on proximity data points. Values missing from the questionare will be replaced with the caluculated values of the random forest imputation algorithim.

The model will be evaluated with and without imputed data points through the resultant R^2 value and the Mean Squared Error (MSE).

```r
#Recode age bands into numeric groupings for an easier interpreation of the median
kneeData["Age Band"][kneeData["Age Band"] == "50 to 59"] <- "1"
kneeData["Age Band"][kneeData["Age Band"] == "60 to 69"] <- "2"
kneeData["Age Band"][kneeData["Age Band"] == "70 to 79"] <- "3"
kneeData["Age Band"][kneeData["Age Band"] == "80 to 89"] <- "4"
kneeData$"Age Band" <- as.numeric(as.character(kneeData$"Age Band"))

#Impute median values into the missing age data
kneeData["Age Band"][is.na(kneeData["Age Band"])] <- median(kneeData$"Age Band", na.rm = TRUE)
kneeData[kneeData == "*"] <- NA

#Call a KNN function designed to split the missing values comparing only to the nearest one neighbour
imputedGender <- kNN(kneeData, k = 1)
kneeData$Gender <- imputedGender$Gender
kneeData$Gender <- as.numeric(kneeData$Gender)

#Functions to replace missing values to specified values
nineToNo <- function(x){
    ifelse(x==9,2,x)
}

#Note: Imputation for the random forest package can only read NA values, all 9 values have to be recode
nineToNA <- function(x){
    ifelse(x==9,NA,x)
}
colsToNo <- c(
    "Heart Disease", "High Bp", "Stroke", "Circulation", "Lung Disease",
    "Diabetes", "Kidney Disease", "Nervous System", "Liver Disease",
     "Cancer", "Depression", "Arthritis"
)

#Function calls to recode missing values
setDT(kneeData)[, (colsToNo) := lapply(.SD, nineToNo), .SDcols=colsToNo]
```

Characteristic	**N = 4,528**[1]
Age Band	
1	164 (3.6%)
2	1,034 (23%)
3	3,114 (69%)
4	216 (4.8%)
Gender	
1	2,019 (45%)
2	2,509 (55%)
Heart Disease	
1	369 (8.1%)
2	4,159 (92%)
Post-Op Q Assisted By	
1	4,484 (99%)
2	18 (0.4%)
3	6 (0.1%)
4	9 (0.2%)
5	2 (<0.1%)
6	9 (0.2%)

[1] n (%)

```
kneeData[, (setdiff(names(kneeData), "Post-Op Q EQ5D Index")) := lapply(.SD, nineToNA),
          .SDcols = setdiff(names(kneeData), "Post-Op Q EQ5D Index")]

#Call function to handle missing values for questionare questions in dataset
kneeData <- na.roughfix(kneeData)

kneeData |> tbl_summary(
    include = c("Age Band", "Gender", "Heart Disease", "Post-Op Q Assisted By"
))
```

```
#Dataset is now split into training and test splits with
#70% of current data in training with 30% reserved for testing
kneeSplit <- kneeData %>%
    initial_split(prop = 0.7, strata = "Post-Op Q EQ5D Index")
    kneeTrain <- training(kneeSplit)
    kneeTest <- testing(kneeSplit)

glimpse(kneeData)
```

```
## Rows: 4,528
## Columns: 77
## $ `Age Band`                                          <dbl> 3, 3, 3, 3, 3, 3, 3~
## $ Gender                                              <dbl> 2, 1, 2, 1, 2, 2, 2~
## $ `Pre-Op Q Assisted`                                 <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Assisted By`                              <dbl> 0, 0, 0, 0, 0, 0, 0~
## $ `Pre-Op Q Symptom Period`                           <dbl> 2, 4, 4, 2, 4, 2, 2~
## $ `Pre-Op Q Previous Surgery`                         <dbl> 2, 1, 2, 2, 2, 2, 2~
```

```
## $ `Pre-Op Q Living Arrangements`                     <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Disability`                              <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Heart Disease`                                    <dbl> 2, 2, 2, 2, 2, 1, 2~
## $ `High Bp`                                          <dbl> 1, 2, 2, 1, 1, 1, 2~
## $ Stroke                                             <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ Circulation                                        <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Lung Disease`                                     <dbl> 2, 2, 2, 1, 2, 2, 2~
## $ Diabetes                                           <dbl> 2, 2, 2, 1, 2, 2, 2~
## $ `Kidney Disease`                                   <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Nervous System`                                   <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Liver Disease`                                    <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ Cancer                                             <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ Depression                                         <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ Arthritis                                          <dbl> 2, 1, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Mobility`                                <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Self-Care`                               <dbl> 1, 1, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Activity`                                <dbl> 3, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Discomfort`                              <dbl> 2, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Anxiety`                                 <dbl> 1, 1, 2, 2, 1, 2, 1~
## $ `Pre-Op Q EQ5D Index Profile`                      <dbl> 21321, 21221, 21332~
## $ `Pre-Op Q EQ5D Index`                              <dbl> 0.364, 0.691, 0.030~
## $ `Post-Op Q Assisted`                               <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Assisted By`                            <dbl> 1, 1, 1, 1, 1, 1, 1~
## $ `Post-Op Q Living Arrangements`                    <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Post-Op Q Disability`                             <dbl> 1, 2, 2, 2, 2, 1, 2~
## $ `Post-Op Q Mobility`                               <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Self-Care`                              <dbl> 2, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Activity`                               <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Discomfort`                             <dbl> 1, 1, 1, 1, 1, 2, 2~
## $ `Post-Op Q Anxiety`                                <dbl> 2, 1, 1, 1, 1, 1, 1~
## $ `Post-Op Q Satisfaction`                           <dbl> 2, 3, 1, 2, 1, 4, 3~
## $ `Post-Op Q Sucess`                                 <dbl> 1, 1, 1, 1, 1, 4, 1~
## $ `Post-Op Q Allergy`                                <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Bleeding`                               <dbl> 2, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Wound`                                  <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Urine`                                  <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Further Surgery`                        <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Readmitted`                             <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q EQ5D Index Profile`                     <dbl> 12112, 11111, 11111~
## $ `Post-Op Q EQ5D Index`                             <dbl> 0.744, 1.000, 1.000~
## $ `Knee Replacement EQ 5D Index Post-Op Q Predicted` <dbl> 0.6621424, 0.740953~
## $ `Pre-Op Q EQ VAS`                                  <dbl> 85, 50, 46, 60, 60,~
## $ `Post-Op Q EQ VAS`                                 <dbl> 60, 100, 90, 95, 90~
## $ `Knee Replacement EQ VAS_Post-Op Q Predicted`      <dbl> 75.63073, 66.38606,~
## $ `Knee Replacement Pre-Op Q Pain`                   <dbl> 1, 1, 0, 0, 1, 1, 1~
## $ `Knee Replacement Pre-Op Q Night Pain`             <dbl> 1, 2, 2, 2, 2, 3, 0~
## $ `Knee Replacement Pre-Op Q Washing`                <dbl> 3, 4, 3, 4, 3, 3, 3~
## $ `Knee Replacement Pre-Op Q Transport`              <dbl> 1, 4, 2, 4, 2, 2, 2~
## $ `Knee Replacement Pre-Op Q Walking`                <dbl> 3, 3, 2, 3, 3, 1, 2~
## $ `Knee Replacement Pre-Op Q Standing`               <dbl> 3, 2, 2, 3, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Limping`                <dbl> 0, 3, 0, 0, 3, 1, 1~
## $ `Knee Replacement Pre-Op Q Kneeling`               <dbl> 0, 0, 0, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Work`                   <dbl> 2, 2, 1, 3, 2, 1, 2~
## $ `Knee Replacement Pre-Op Q Confidence`             <dbl> 3, 3, 3, 2, 1, 1, 2~
```

```
## $ `Knee Replacement Pre-Op Q Shopping`             <dbl> 2, 4, 0, 2, 3, 0, 2~
## $ `Knee Replacement Pre-Op Q Stairs`               <dbl> 1, 4, 2, 2, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Score`                <dbl> 20, 32, 17, 27, 27,~
## $ `Knee Replacement Post-Op Q Pain`                <dbl> 4, 3, 3, 4, 3, 1, 2~
## $ `Knee Replacement Post-Op Q Night Pain`          <dbl> 4, 4, 3, 4, 4, 3, 2~
## $ `Knee Replacement Post-Op Q Washing`             <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Transport`           <dbl> 2, 4, 4, 4, 4, 3, 4~
## $ `Knee Replacement Post-Op Q Walking`             <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Standing`            <dbl> 3, 4, 4, 4, 4, 1, 3~
## $ `Knee Replacement Post-Op Q Limping`             <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Kneeling`            <dbl> 0, 4, 0, 3, 2, 0, 2~
## $ `Knee Replacement Post-Op Q Work`                <dbl> 3, 3, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Confidence`          <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Shopping`            <dbl> 4, 4, 4, 4, 4, 0, 4~
## $ `Knee Replacement Post-Op Q Stairs`              <dbl> 3, 4, 4, 4, 3, 1, 4~
## $ `Knee Replacement Post-Op Q Score`               <dbl> 37, 46, 42, 47, 44,~
## $ `Knee Replacement OKS Post-Op Q Predicted`       <dbl> 34.02619, 36.88715,~
```

```r
print(colnames(kneeData))
```

```
##  [1] "Age Band"
##  [2] "Gender"
##  [3] "Pre-Op Q Assisted"
##  [4] "Pre-Op Q Assisted By"
##  [5] "Pre-Op Q Symptom Period"
##  [6] "Pre-Op Q Previous Surgery"
##  [7] "Pre-Op Q Living Arrangements"
##  [8] "Pre-Op Q Disability"
##  [9] "Heart Disease"
## [10] "High Bp"
## [11] "Stroke"
## [12] "Circulation"
## [13] "Lung Disease"
## [14] "Diabetes"
## [15] "Kidney Disease"
## [16] "Nervous System"
## [17] "Liver Disease"
## [18] "Cancer"
## [19] "Depression"
## [20] "Arthritis"
## [21] "Pre-Op Q Mobility"
## [22] "Pre-Op Q Self-Care"
## [23] "Pre-Op Q Activity"
## [24] "Pre-Op Q Discomfort"
## [25] "Pre-Op Q Anxiety"
## [26] "Pre-Op Q EQ5D Index Profile"
## [27] "Pre-Op Q EQ5D Index"
## [28] "Post-Op Q Assisted"
## [29] "Post-Op Q Assisted By"
## [30] "Post-Op Q Living Arrangements"
## [31] "Post-Op Q Disability"
## [32] "Post-Op Q Mobility"
## [33] "Post-Op Q Self-Care"
## [34] "Post-Op Q Activity"
## [35] "Post-Op Q Discomfort"
```

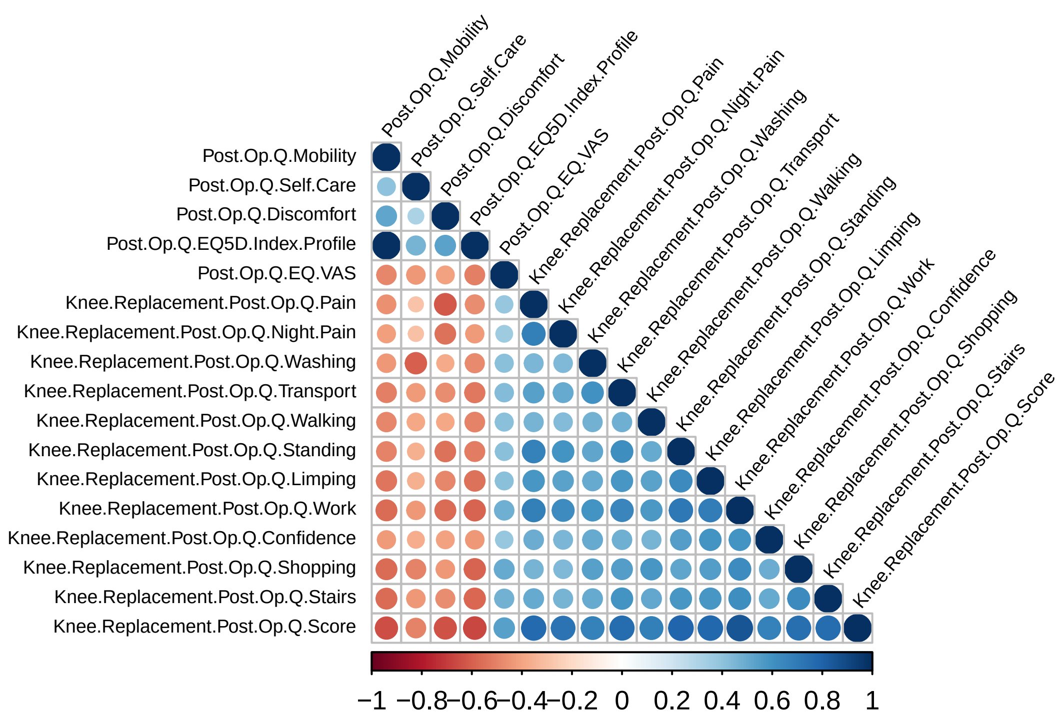
```
## [36] "Post-Op Q Anxiety"
## [37] "Post-Op Q Satisfaction"
## [38] "Post-Op Q Sucess"
## [39] "Post-Op Q Allergy"
## [40] "Post-Op Q Bleeding"
## [41] "Post-Op Q Wound"
## [42] "Post-Op Q Urine"
## [43] "Post-Op Q Further Surgery"
## [44] "Post-Op Q Readmitted"
## [45] "Post-Op Q EQ5D Index Profile"
## [46] "Post-Op Q EQ5D Index"
## [47] "Knee Replacement EQ 5D Index Post-Op Q Predicted"
## [48] "Pre-Op Q EQ VAS"
## [49] "Post-Op Q EQ VAS"
## [50] "Knee Replacement EQ VAS_Post-Op Q Predicted"
## [51] "Knee Replacement Pre-Op Q Pain"
## [52] "Knee Replacement Pre-Op Q Night Pain"
## [53] "Knee Replacement Pre-Op Q Washing"
## [54] "Knee Replacement Pre-Op Q Transport"
## [55] "Knee Replacement Pre-Op Q Walking"
## [56] "Knee Replacement Pre-Op Q Standing"
## [57] "Knee Replacement Pre-Op Q Limping"
## [58] "Knee Replacement Pre-Op Q Kneeling"
## [59] "Knee Replacement Pre-Op Q Work"
## [60] "Knee Replacement Pre-Op Q Confidence"
## [61] "Knee Replacement Pre-Op Q Shopping"
## [62] "Knee Replacement Pre-Op Q Stairs"
## [63] "Knee Replacement Pre-Op Q Score"
## [64] "Knee Replacement Post-Op Q Pain"
## [65] "Knee Replacement Post-Op Q Night Pain"
## [66] "Knee Replacement Post-Op Q Washing"
## [67] "Knee Replacement Post-Op Q Transport"
## [68] "Knee Replacement Post-Op Q Walking"
## [69] "Knee Replacement Post-Op Q Standing"
## [70] "Knee Replacement Post-Op Q Limping"
## [71] "Knee Replacement Post-Op Q Kneeling"
## [72] "Knee Replacement Post-Op Q Work"
## [73] "Knee Replacement Post-Op Q Confidence"
## [74] "Knee Replacement Post-Op Q Shopping"
## [75] "Knee Replacement Post-Op Q Stairs"
## [76] "Knee Replacement Post-Op Q Score"
## [77] "Knee Replacement OKS Post-Op Q Predicted"
```

2. Selection and preprocessing of predictors

The first step to select important predictors to the model is to preform correlation analysis to identify predictors. There are 77 possible predictors in this dataset visually representing all of the possible predictor variables made the correlation plot difficult to interpret, therefore a selection of predictor variables have been selected from the complete correlation plot (ref Appendix) only if they had a value +/- 0.5 associated with any of the variables. This is a purely numeric based approach, however, there may be certain variables that hold clinical significance and wish to be inlcuded in the model to evaluate ther relationship.

```r
#Trailing spaces were present in some column names, these needed to be stripped.
#All spaces in the columns were replaced with a . character
colnames(kneeTrain) <- make.names(colnames(kneeTrain))
```

```r
colnames(kneeTest) <- make.names(colnames(kneeTest))

kneeCorColumns <- kneeTrain[, c(
    "Post.Op.Q.Mobility", "Post.Op.Q.Self.Care", "Post.Op.Q.Discomfort",
    "Post.Op.Q.EQ5D.Index.Profile","Post.Op.Q.EQ.VAS", "Knee.Replacement.Post.Op.Q.Pain",
    "Knee.Replacement.Post.Op.Q.Night.Pain","Knee.Replacement.Post.Op.Q.Washing",
    "Knee.Replacement.Post.Op.Q.Transport", "Knee.Replacement.Post.Op.Q.Walking",
    "Knee.Replacement.Post.Op.Q.Standing", "Knee.Replacement.Post.Op.Q.Limping",
    "Knee.Replacement.Post.Op.Q.Work","Knee.Replacement.Post.Op.Q.Confidence",
    "Knee.Replacement.Post.Op.Q.Shopping", "Knee.Replacement.Post.Op.Q.Stairs",
    "Knee.Replacement.Post.Op.Q.Score"
)]

kneeCorr <- cor(kneeCorColumns)
corrplot(kneeCorr,
    type = "lower", tl.srt = 50,
    tl.col = "black", number.cex = 0.5, tl.cex = 0.6
)
```



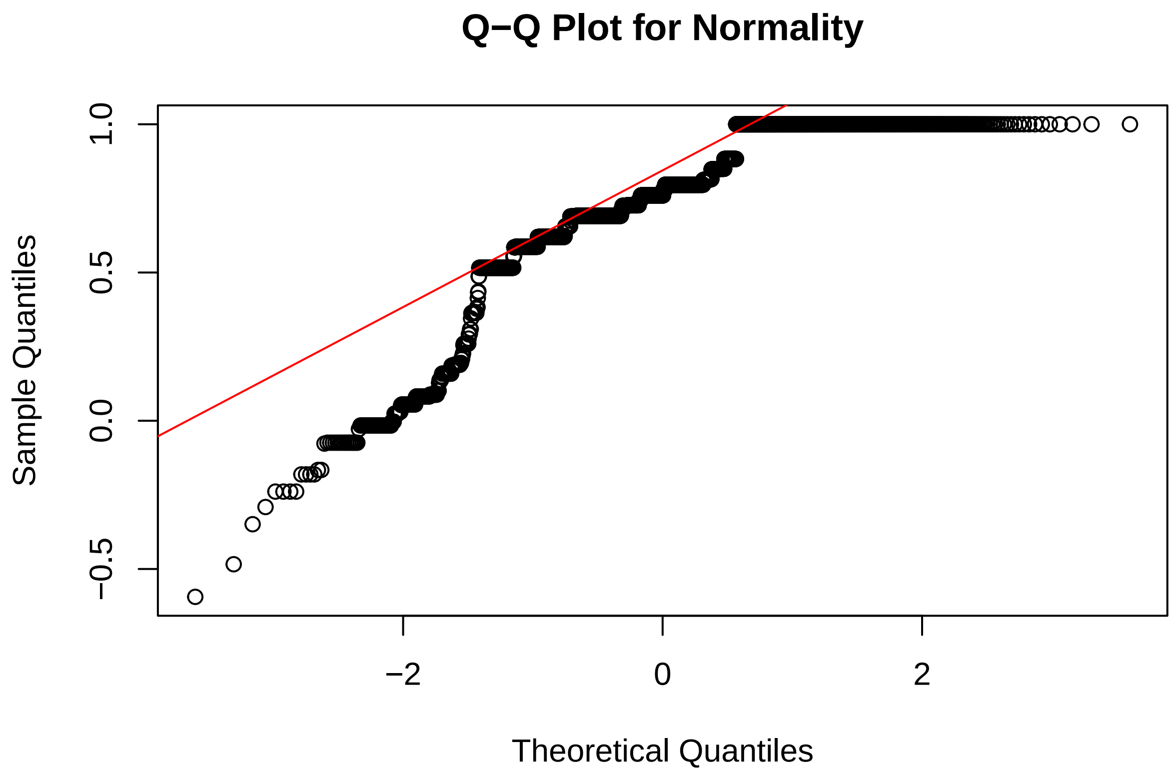

```r
# Create a recipie with signficant predictor variables
# Recipie used for more readable code when fitting the model, assignment specificatoins mention not fit
# In the case of multiple models being fitted this recipie can be used multiple times to the parameter

kneeModelRec <- kneeTrain %>%
    recipe(Post.Op.Q.EQ5D.Index ~
        Post.Op.Q.Mobility +
```

```r
        Post.Op.Q.Self.Care +
        Post.Op.Q.Discomfort +
        Post.Op.Q.EQ5D.Index.Profile +
        Post.Op.Q.EQ.VAS +
        Knee.Replacement.Post.Op.Q.Pain +
        Knee.Replacement.Post.Op.Q.Night.Pain +
        Knee.Replacement.Post.Op.Q.Washing +
        Knee.Replacement.Post.Op.Q.Transport +
        Knee.Replacement.Post.Op.Q.Walking +
        Knee.Replacement.Post.Op.Q.Standing +
        Knee.Replacement.Post.Op.Q.Limping +
        Knee.Replacement.Post.Op.Q.Work +
        Knee.Replacement.Post.Op.Q.Confidence +
        Knee.Replacement.Post.Op.Q.Shopping +
        Knee.Replacement.Post.Op.Q.Stairs +
        Knee.Replacement.Post.Op.Q.Score) %>%
    step_zv(all_predictors()) %>%
    step_corr(all_predictors())
```

3. Model specification and training

Missing values in the dataset have been addressed alongside idenifying important predictor variables that have influence on the dependent variable. There still lies the issue of using a linear regression model or a random forest. Due to its simpler computation and interpratibility linear models are preffered but a series of assumptions need to hold true, in order to use this model. The following code runs a simple Q-Q test that checks for normality within the data.

```r
qqnorm(kneeTrain$Post.Op.Q.EQ5D.Index, main = "Q-Q Plot for Normality")
qqline(kneeTrain$Post.Op.Q.EQ5D.Index, col="red")
```



As shown in the Q-Q plot, the data in terms of the dependent variable does not hold the assumption of normality. Since this assumption is unmet, a random forest plot will be used. Random forest plots are capable of handiling regression problems with non-linear properties.

Calling the Random Forest Model

The following code depicts the RF model being called for the purposes of fitting a regression model. RF works on the basis of ensemble learning, where multiple decision trees are used to reach a consensus on the. In this problem specification the tree first starts off with the full complete training dataset and creates multiple branches of decision trees, giving each decision tree its own sample size (with replacement) of the original training dataset. This process is repeated for every decision tree in the RF as shown in the workflow below. These decision trees all contain a unique subset of the training data and reach a predicted value for the Post EQ5 Index based on their own subset of the data. It's important to note that each subset may not have the same types of significant variables as found in the correlation plot when predicting the value. This is an important feature, as this is how the model determines which variables are important to the model by comparing the accuracy of the trees that contain certain variables with the actual value of the Post EQ5 Index To ensure that the trees have a collective prediction that is accurate, they go through a process known as bootstrapping designed to average out the predicted values across all the different trees. There is an optimal number of decision trees to consider that will decrease the overall variance while maintaining computational efficiency, but, having a larger number of trees doesn't necessarily mean increased model performance. The model reaches a point where the increase of computation does not justify the marginal increase in the predicted value. This problem arises due to overfitting within the model, where the RF starts to learn from noise in the data, rather than making generalisable predictions based on the data. To find the optimal number of trees the model would have to go through hyper tuning tests which is outside the scope of this report, therefore the default value of 500 decision trees built in within the randomForest package will be used.

```r
#Initate random forest model
kneeRandForestNoImportance <- rand_forest() %>%
    set_mode("regression") %>%
    set_engine("randomForest")

# Use recipie against specifications of model
kneeWorkflowNoImp <- workflow() %>%
    add_recipe(kneeModelRec) %>%
    add_model(kneeRandForestNoImportance)

# Fit model with training data
kneeFitModelNoImp <- kneeWorkflowNoImp %>%
    fit(data = kneeTrain)

#Specify a model with built in importance feature for random forest
kneeRandForest <- rand_forest() %>%
    set_mode("regression") %>%
    set_engine("randomForest", importance = TRUE)

kneeWorkflow <- workflow() %>%
    add_recipe(kneeModelRec) %>%
    add_model(kneeRandForest)


extractRF <-
kneeFitModel <- kneeWorkflow %>%
    fit(data = kneeTrain)
```

Identifying Influential Variables

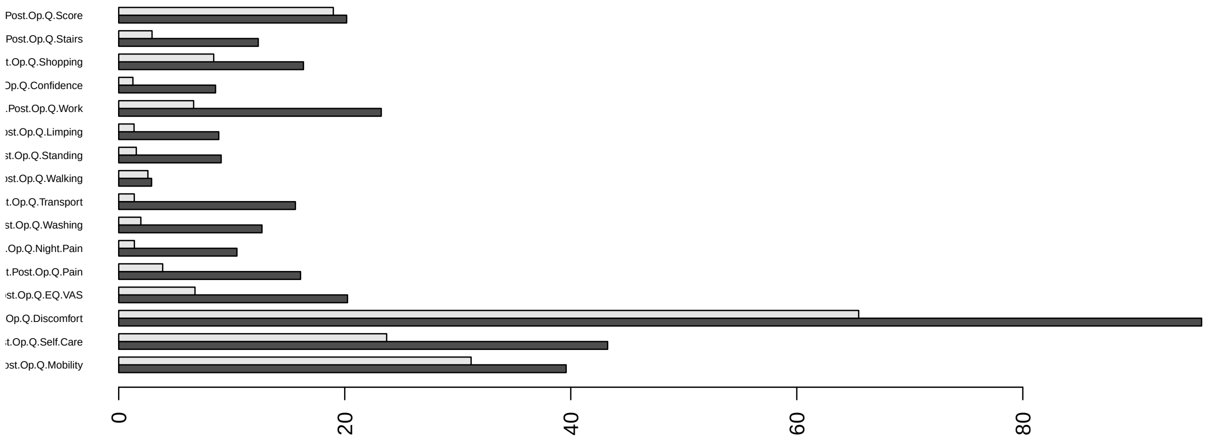
Out of the 16 possible significant variables, only 4 showed a degree of importance. In this report the threshold for importance was categorised as any value of importance greater than 20 units. The way in which the vip package places importance on variables is significant to our understanding of how variables affect the model and how they are important to the final Post EQ5 Index. The vip pacakage uses a method known as Out-Of-Bag (OOB) to create validation sets errors for each descion tree. This means that the each predictor in the validation set is shuffled around each of the trees and the changes in the validation set errors (VSE) for each of the trees are recorded, the trees that exhibit a great change in the VSE errors on the change of a predictor in the sample set, the value is noted and the averaged imporvment of the VSE upon the addition of a certain variable is recorded. The documentation notes certain features that imapct the validity of the results, such as: high correlation in the variables, varying scales of measurement for the predictor variables and larger sample sizes resuting in more accurate VSE scores. In the context of this report, there were some issues encountered that directly affect the way the data interacts with the vip package. Upon visual inspection there was a certain predictor "Knee Replacement Post Op Q Score" that would change between +/5 units in the importance score, this was significant due to the threshold of an important varibale being 20 units and depending on the specific run, that specific variable was considered important. Knee Replacement Post OP Q score can be assumed to be correlated to the Post Op Q Pain score, looking at the PROMS data dictionary, it states

"Each of the KR question has 5 possible responses which each are attributable 0-4 points. The KR score is derived by adding the points for each question. The maximum number of points is 48 which would signify the best outcome, the minimum being 0."

Since the KR Post Op Q score is an agrregate variable it will change depending on the specific variables included in the sample sizes of the descion tress in the RF model.

```r
kneeFitModelNoImp %>%
    extract_fit_parsnip() %>%
    vip(num_features=5)
```


```r
# Trained RF model needs to be extracted from tidymodels workflow in order to use it within varImpPlot
extractRF <- extract_fit_parsnip(kneeFitModel)
MeanImportanceValues <- importance(extractRF$fit)
barplot(
    t(MeanImportanceValues),
    beside = TRUE,
    names.arg=rownames(MeanImportanceValues),
    las=2,
    cex.names=0.5,
    horiz=TRUE

)
```

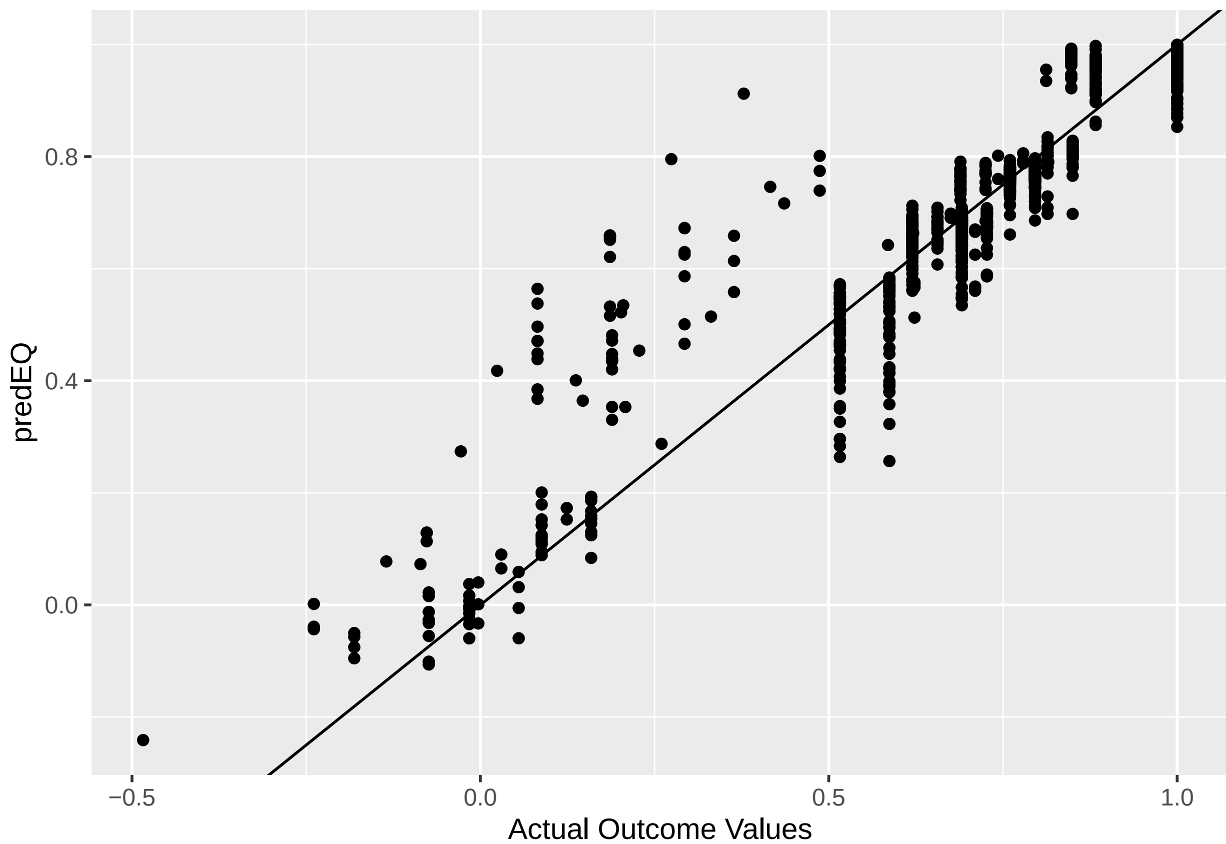


4. Model evaluation Code has

```
predEQVal <- kneeFitModel %>%
    predict(new_data = kneeTest)

kneeResult <- kneeTest %>%
    bind_cols(predEQVal) %>%
    rename(predEQ = .pred)

metrics(kneeResult, truth = Post.Op.Q.EQ5D.Index, estimate = predEQ)
```

```
## # A tibble: 3 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>          <dbl>
## 1 rmse    standard      0.0842
## 2 rsq     standard      0.886
## 3 mae     standard      0.0477
```

```
kneeResult %>%
    ggplot(aes(x = Post.Op.Q.EQ5D.Index, y = predEQ)) +
    geom_abline(intercept = 0, slope = 1) +
    geom_point() +
    xlab("Actual Outcome Values")
```




```
    ylab("Predicted Outcome Values")
```

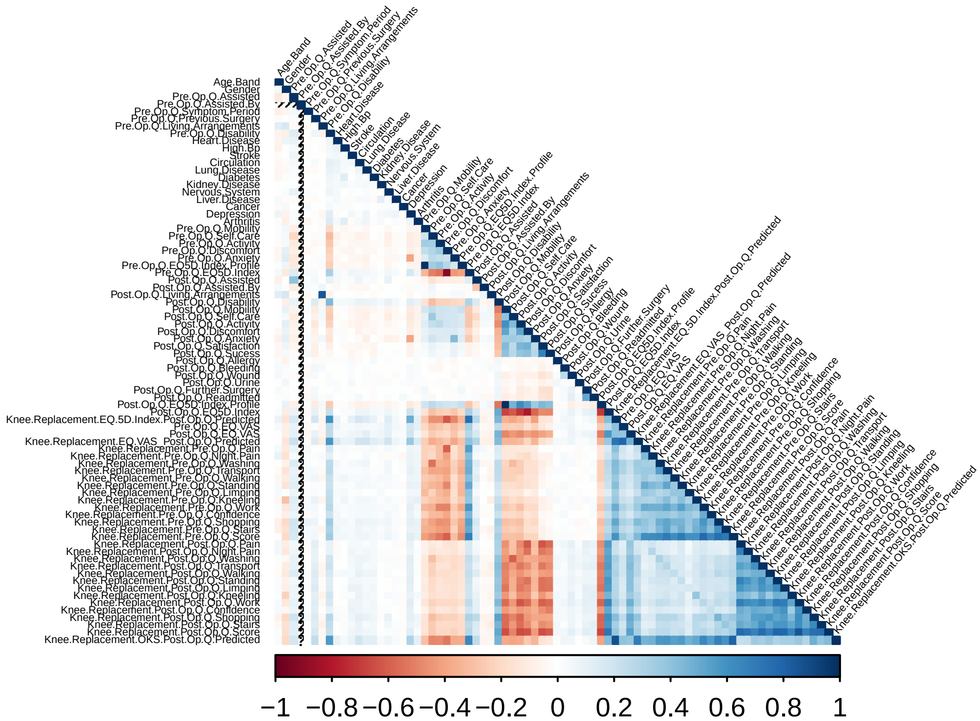
```
## $y
## [1] "Predicted Outcome Values"
##
## attr(,"class")
## [1] "labels"
```

Limitations of machine learning model

Real problem with the number of trees being used in the model, the number of descion trees is simply not enough as the VSE error for certain predictors keep changing by +/- 5% which is significant when you have some variables within the threshold.

Appendix

```
kneeCorr <- cor(kneeTrain)
corrplot(kneeCorr, method = "color", type = "lower", tl.col = "black", tl.srt = 50, number.cex = 0.5, tl
```


Age.Band
Gender
Pre.Op.Q.Assisted
Pre.Op.Q.Assisted.By
Pre.Op.Q.Symptom.Period
Pre.Op.Q.Previous.Surgery
Pre.Op.Q.Living.Arrangements
Pre.Op.Q.Disability
Heart.Disease
High.Bp
Stroke
Circulation
Lung.Disease
Diabetes
Kidney.Disease
Nervous.System
Liver.Disease
Cancer
Depression
Arthritis
Pre.Op.Q.Mobility
Pre.Op.Q.Self.Care
Pre.Op.Q.Activity
Pre.Op.Q.Discomfort
Pre.Op.Q.Anxiety
Pre.Op.Q.EQ5D.Index.Profile
Pre.Op.Q.EQ5D.Index
Post.Op.Q.Assisted
Post.Op.Q.Assisted.By
Post.Op.Q.Living.Arrangements
Post.Op.Q.Disability
Post.Op.Q.Mobility
Post.Op.Q.Self.Care
Post.Op.Q.Activity
Post.Op.Q.Discomfort
Post.Op.Q.Anxiety
Post.Op.Q.Satisfaction
Post.Op.Q.Sucess
Post.Op.Q.Allergy
Post.Op.Q.Bleeding
Post.Op.Q.Wound
Post.Op.Q.Urine
Post.Op.Q.Further.Surgery
Post.Op.Q.Readmitted
Post.Op.Q.EQ5D.Index.Profile
Post.Op.Q.EQ5D.Index
Knee.Replacement.EQ.5D.Index.Post.Op.Q.Predicted
Pre.Op.Q.EQ.VAS
Post.Op.Q.EQ.VAS
Knee.Replacement.EQ.VAS_Post.Op.Q.Predicted
Knee.Replacement.Pre.Op.Q.Pain
Knee.Replacement.Pre.Op.Q.Night.Pain
Knee.Replacement.Pre.Op.Q.Washing
Knee.Replacement.Pre.Op.Q.Transport
Knee.Replacement.Pre.Op.Q.Walking
Knee.Replacement.Pre.Op.Q.Standing
Knee.Replacement.Pre.Op.Q.Limping
Knee.Replacement.Pre.Op.Q.Kneeling
Knee.Replacement.Pre.Op.Q.Work
Knee.Replacement.Pre.Op.Q.Confidence
Knee.Replacement.Pre.Op.Q.Shopping
Knee.Replacement.Pre.Op.Q.Stairs
Knee.Replacement.Pre.Op.Q.Score
Knee.Replacement.Post.Op.Q.Pain
Knee.Replacement.Post.Op.Q.Night.Pain
Knee.Replacement.Post.Op.Q.Washing
Knee.Replacement.Post.Op.Q.Transport
Knee.Replacement.Post.Op.Q.Walking
Knee.Replacement.Post.Op.Q.Standing
Knee.Replacement.Post.Op.Q.Limping
Knee.Replacement.Post.Op.Q.Kneeling
Knee.Replacement.Post.Op.Q.Work
Knee.Replacement.Post.Op.Q.Confidence
Knee.Replacement.Post.Op.Q.Shopping
Knee.Replacement.Post.Op.Q.Stairs
Knee.Replacement.Post.Op.Q.Score
Knee.Replacement.OKS.Post.Op.Q.Predicted
−1 −0.8 −0.6 −0.4 −0.2 0 0.2 0.4 0.6 0.8 1